SARTHAK THETE

ENTC Student | Aspiring Embedded Systems Engineer

7796532352 sarthak.thete@mitwpu.edu.in sarthak-thete-52092828 Pune, India

PROFESSIONAL SUMMARY

ECE engineering student focused on embedded systems and digital design. Hands-on experience with technical projects, C++, MATLAB. Passionate about building practical hardware solutions and working in collaborative environments.

EDUCATIONS

MIT WPU | Pune — Aug 2023 - Present
Bachelor of Technology in Electronics and Communication Engineering — *CGPA : 7.19 CGPA*

PROJECTS

Vision-Based Smart Checkout System (ZeroWait.store) | Site — Feb 2026 - Apr 2026
Embedded & IoT Developer
Developed a vision-based smart checkout system utilizing edge AI and IoT for automated billing.
- Established an object detection pipeline on Raspberry Pi 5, achieving ~85-90% accuracy.
- Trained a custom model with Roboflow (SAM-based annotation), enhancing detection performance by ~35%.
- Integrated ESP32 and HX711 via MQTT for real-time data transfer.
- Implemented sensor fusion (vision + weight), increasing reliability by ~40%.
- Reduced checkout time by ~70%.
- Currently preparing a research paper on system architecture and performance analysis.

Technologies /Tool Used : *Computer Vision, IoT, AI/ML*

Arduino-Based CNC Writing Machine using GRBL | Github — Jan 2026 - Mar 2026
Embedded Systems Developer / Hardware Developer
Developed a 2-axis CNC writing machine using Arduino and GRBL firmware for automated plotting.
- Achieved 90% writing accuracy through calibrated stepper motors and CNC shield.
- Integrated X-Y motion control with a servo-based pen mechanism.
- Utilized Universal G-code Sender and Inkscape for converting designs into G-code and executing operations.
- Enhanced system efficiency by 60% through optimized calibration and workflow.

Technologies /Tool Used : *G-code Processing, CNC Systems, Stepper Motor Control, Serial Communication*

Mini gaming Console — Oct 2024 - Oct 2024
Embedded System Project
- Developed a mini gaming console using ESP32 Microcontroller.
- Configured and programmed the system using C for basic control and interfacing tasks(low level coding).
- Interfaced **4 GPIO input buttons** for real-time user control.
- Implemented and tested classic games such as Mario, Kirby, and Pokémon FireRed.
- Achieved system response time of **under 1 second** during gameplay.

Technologies /Tool Used : *ESP32, C, GPIO, Embedded Systems, Button Interfacing*

SKILLS

Microcontrollers/Boards :	STM32(ARM Cortex-M), ESP32, Arduino(ATmega328P), C8051
Languages :	English, Hindi, Marathi, German, French
Programming Languages :	C, Embedded C, Python (Basic), Data Structures in C
Soft Skills :	Communication, Teamwork, Problem Solving, Adaptability
Tools Platforms :	STM32CubeIDE, Keil uVision, Arduino IDE, MATLAB, Simulink, Serial Debugging tools

EXPERIENCE

- National Academic Immersion Program 2025 (iACE), Gandhinagar: Hands-on practical training on real industrial machines and systems, covering Automotive Communication Protocols(CAN & LIN - Fundamentals), ADAS concepts, industrial automation, and Industry 4.0 workflows; included on-site industrial exposure at Ratnamani Metals & Tubes Ltd.

CERTIFICATIONS

Industrial Training Programme - Artificial Intelligence & Machine Learning — Jul 2025
international Automobile Centre of Excellence (iACE-GUJARAT)